Results, Storyline, Findings, and Conclusions for ML Papers

2-page handout for CS / ML students writing conference papers

1. What the Results section is for

The Results section is where the paper makes its empirical case. It should show what was observed, whether the evidence supports the claims, how reliable the effect is, and where the limits are. A good Results section is not a data dump; it is a structured argument.

2. Results vs findings vs conclusions

Part	Main question	Typical content
Results	What did we observe?	Tables, figures, quantitative patterns.
Findings / Discussion	What do the results mean?	Interpretation, mechanism, implications.
Conclusion	What should the reader remember?	Take-home message, contribution, limits, next step.

Teaching rule: **Results = evidence**; **Findings = interpretation**; **Conclusion = final takeaway**.

3. Organize results around claims

Claim	Evidence
Our method performs better	Main benchmark table.
Our method is more stable	Learning curves or variance plot.
Component X matters	Ablation table.
The method scales or is robust	Performance versus agents, graph size, or perturbation.

4. A good ML results storyline

- **Headline result** — does the method work on the main task?
- **Breadth** — does it work across settings?
- **Explanation** — what do ablations show?
- **Reliability** — what do seeds, variance, and robustness show?
- **Limits** — where does the effect weaken or disappear?

5. How to write a results paragraph

A strong results paragraph usually has four moves: point to the figure or table, state the main pattern, interpret it, and qualify it if needed.

Template: Table/Figure X shows that *[main pattern]*. Compared with *[baseline]*, *[method]* improves *[metric]* by *[amount]*. This suggests that *[interpretation]*. The effect is strongest in *[condition]*, but weaker in *[condition]*.

6. Tables vs figures

- **Use tables** when exact values and direct comparison matter.
- **Use figures** when trends, learning dynamics, distributions, or uncertainty matter.
- **Typical pairing:** table for final benchmark comparison, figure for learning curves, table for ablations, figure for robustness or scaling trends.

7. Report uncertainty honestly

Good ML results usually report multiple seeds and mean with standard deviation or confidence intervals. Do not report only the best run. If the method is unstable, say so.

8. From results to findings

A **result** is an observed outcome. A **finding** is a broader conclusion supported by several results.

Type	Example
Result	Communication improves return by 8–12% in large networks.
Finding	Communication is most useful in large, congested settings where decentralized agents strongly affect one another.

Synthesis template: Taken together, the results indicate that *[main finding]*. This is supported by *[result 1]*, *[result 2]*, and *[result 3]*. The effect is strongest under *[condition]* and weaker under *[condition]*.

9. From findings to conclusions

A conclusion should not repeat every number. It should briefly state what the paper showed, why it matters, the main limitation, and what should happen next.

- **Restate the problem** or goal.
- **State the main finding.**
- **State the contribution or implication.**
- **Note one key limitation** or future direction.

Conclusion template: This paper studied *[problem]* and showed that *[main finding]*. These results suggest that *[implication]*. However, the study is limited by *[limitation]*. Future work should examine *[next step]*.

10. Common mistakes

Mistake	Why it is weak
Too many tables, no story	The reader cannot tell what matters.
Reporting only best runs	Variance and instability are hidden.
Reading numbers row by row	There is no synthesis or interpretation.
Hiding weak cases	Trust is reduced.
Overclaiming in conclusions	The conclusion goes beyond the evidence.

11. Fast checklist

- My Results section is organized around claims.
- The headline result appears early.
- Each figure or table supports one clear point.
- My text interprets patterns, not just numbers.
- I report uncertainty where needed.
- I mention weak or limiting cases honestly.
- My findings are broader than single results, but still evidence-based.
- My conclusion states the takeaway, contribution, and limit in a few sentences.

12. In-class exercise

Main claim: Our paper claims that ...
Headline result: The main evidence is shown in ...
Main finding: Taken together, the results indicate that ...
Conclusion: This paper shows that ... This matters because ... A limitation is ... Next, we should ...